



Custom Power and Control PCB for the RoCam Gimbal Tracker

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March 2026

Revision History

Date	Version	Notes
2026-03-04	1.0	Initial design documentation

Contents

1	Symbols, Abbreviations, and Acronyms	iii
2	Introduction	1
3	Design Constraints	1
3.1	Power Budget	1
3.2	Input Power	1
3.3	Voltage Rails	2
3.4	Environmental and Safety Constraints	2
3.5	Budget Constraints	2
4	System Block Diagram	3
5	Component Selection	3
5.1	Buck Converter: TPS54308	3
5.2	USB PD Controller: HUSB237	4
5.3	Microcontroller: STM32F405RG1x	4
5.4	Inductor: FXL0630-15q-M	4
5.5	Connectors	5
6	Circuit Design and Calculations	5
6.1	USB-C Power Delivery	5
6.2	Servo Power Regulators	6
6.2.1	Output Voltage Calculation	7
6.2.2	Component Selection	8
6.3	Jetson Power Regulator	8
6.3.1	Output Voltage Calculation	9
6.3.2	Enable Circuit	9
6.4	STM32 Microcontroller	9
6.4.1	Decoupling and Power	10
6.4.2	Clock Source	10
6.4.3	Status Indicators	11
6.5	Communication Interfaces	11
6.5.1	UART to Jetson	11
6.5.2	PWM to Servos	11
7	PCB Layout	12
7.1	Layout Considerations	14
8	Conclusion	14
A	Full Schematics	16
B	Bill of Materials	19

1 Symbols, Abbreviations, and Acronyms

Symbol	Description
PCB	Printed Circuit Board
PD	Power Delivery (USB-C specification)
UART	Universal Asynchronous Receiver-Transmitter
PWM	Pulse Width Modulation
FCCM	Forced Continuous Conduction Mode
SWD	Serial Wire Debug
CRC	Cyclic Redundancy Check
MCU	Microcontroller Unit
TPS54308	Texas Instruments 3A synchronous buck converter
HUSB237	Hynetek USB PD sink controller
STM32F405	STMicroelectronics ARM Cortex-M4 microcontroller
Jetson	NVIDIA Jetson Orin Nano embedded GPU module
V_{ref}	Internal voltage reference (typically 0.6V for TPS54308)
V_{out}	Regulated output voltage
μF	Microfarad (capacitance unit)
mA	Milliampere (current unit)
MHz	Megahertz (frequency unit)

2 Introduction

The RoCam (High Performance Vision-Guided Rocket Tracker) is a system developed for the McMaster Rocketry Team to autonomously track and record model rocket launches using a gimbal-mounted camera. The system must operate in outdoor environments, track targets with apogees up to 200 meters, and provide stabilized video footage superior to manual camera operation.

This report documents the custom Printed Circuit Board (PCB) designed to support the RoCam system. The PCB, designated “Tracking Cam Prototype 0.5,” integrates three critical functions into a single compact design:

1. **Jetson Orin Nano Power Delivery:** A dedicated 17V, 2A regulator provides stable power to the NVIDIA Jetson Orin Nano, which runs the vision processing and tracking pipeline.
2. **Servo Power Delivery:** Two independent 5.5V, 2A regulators power the pan and tilt servos that control the gimbal.
3. **Gimbal Control:** An STM32F405 microcontroller manages real-time servo control, communicates with the Jetson via UART, and implements the gimbal protocol defined by the McMaster Rocketry Team.

The design addresses the hardware constraints specified in the Software Requirements Specification (SRS) for the RoCam project ([Smith and Koothoor, 2016](#)), including the requirement for an STM32-family microcontroller (SC-2) and NVIDIA Jetson Orin Nano (SC-3) as the compute platform. The board operates from a single USB-C Power Delivery (PD) source, enabling fully portable field deployment without requiring AC power infrastructure.

3 Design Constraints

The PCB design was constrained by system requirements, environmental conditions, safety considerations, and budget limitations derived from the project specification.

3.1 Power Budget

The total power budget is derived from the USB-C PD input capability and the loads’ requirements:

The 65W USB-C PD input provides sufficient headroom for conversion inefficiencies and transient loads.

3.2 Input Power

The board accepts power from a single USB-C PD source capable of delivering 20V at 3.25A (65W). This configuration was chosen for field portability; operators can use any USB-C PD charger or power bank capable of 65W output, eliminating the need for AC power at the launch site (addressing SRS requirement AS-5).

Table 1: Power budget allocation for the RoCam PCB.

Load	Voltage	Current	Power
Jetson Orin Nano	17V	2.0A	34W
Pan servo	5.5V	2.0A	11W
Tilt servo	5.5V	2.0A	11W
STM32F405	3.3V	0.1A	0.33W
Total			56.33W
Input (PD)	20V	3.25A	65W
Headroom			8.67W (13%)

3.3 Voltage Rails

The PCB generates three regulated voltage rails:

- **17V rail:** Powers the Jetson Orin Nano through a DC barrel connector. The 17V output stays within the Jetson’s specified 7–20V input range while providing margin above the minimum to account for cable losses and transients.
- **5.5V rails:** Two independent rails for the pan and tilt servos. The 5.5V output (nominally 5.6V) provides compatibility with standard hobby servos while remaining below their typical 6V maximum rating.
- **3.3V rail:** Derived from the Jetson’s 3.3V output on the expansion header to power the STM32F405 microcontroller.

3.4 Environmental and Safety Constraints

- **Operating temperature:** 0°C to 45°C ambient (SRS EPE-1), with Jetson internal temperatures not exceeding 80°C during 30-minute continuous operation.
- **E-stop functionality:** While the current prototype does not include the planned emergency stop switch (to be added in a future revision), the power architecture supports a high-side cut-off for safe shutdown.
- **Form factor:** The PCB dimensions (approximately 100mm × 80mm) allow mounting on or near the gimbal assembly without interfering with range of motion.

3.5 Budget Constraints

The total project hardware budget is limited to \$500 CAD (SRS BC-1). The PCB and components must fit within this allocation alongside the Jetson, camera, servos, and gimbal hardware.

4 System Block Diagram

Figure 1 presents the system-level block diagram showing power flow (black arrows) and signal flow (blue arrows).

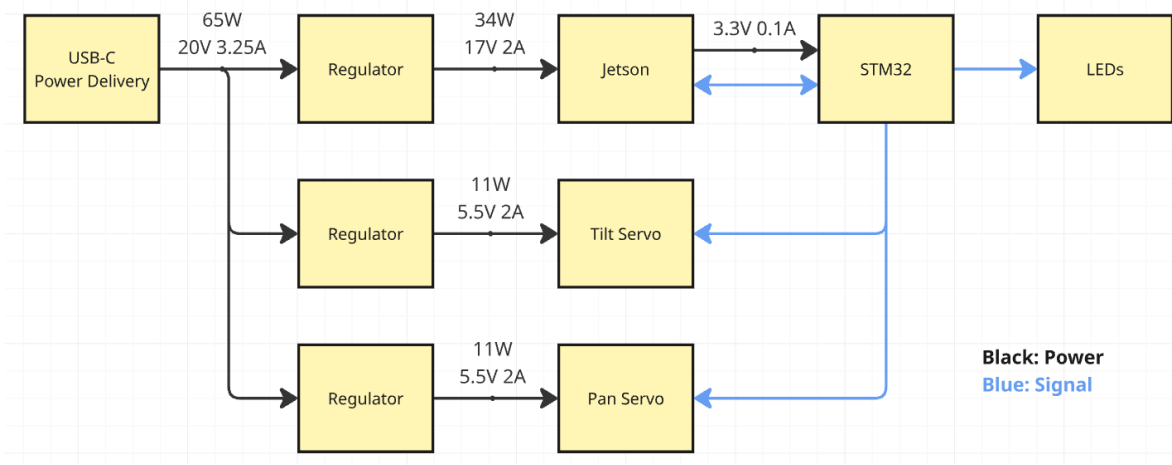


Figure 1: System block diagram showing power delivery from USB-C PD through three independent buck converters to the Jetson (17V), tilt servo (5.5V), and pan servo (5.5V). The STM32F405 receives 3.3V from the Jetson header and communicates via UART while generating PWM signals for servo control. Black arrows indicate power flow; blue arrows indicate signal flow.

The USB-C connector accepts 20V/3.25A from a PD-compliant source. The HUSB237 PD controller negotiates the 20V contract with the source. Three independent TPS54308 buck converters step down the 20V input to the required output voltages. The STM32F405 interfaces with the Jetson via UART (PA9/PA10) for command and telemetry exchange, and drives the servos via software-generated PWM signals on PB5 (tilt) and PB6 (pan).

5 Component Selection

5.1 Buck Converter: TPS54308

The Texas Instruments TPS54308 was selected for all three DC-DC converters based on the following criteria:

- **Input voltage range:** 4.5V to 28V covers the 20V PD input with margin for transients.
- **Output current:** 3A maximum continuous output exceeds the 2A requirements for both the Jetson and servo rails.
- **Package:** SOT-23-6 enables compact layout and adequate thermal performance for the power levels involved.

- **Switching frequency:** Fixed 350kHz with Forced Continuous Conduction Mode (FCCM) provides predictable noise characteristics.
- **Internal compensation:** Eliminates external compensation components, reducing BOM complexity and board area.

The TPS54308 features an internal 0.6V voltage reference (V_{ref}) used in the feedback resistor divider to set the output voltage ([Texas Instruments, 2021](#)).

5.2 USB PD Controller: HUSB237

The Hynetek HUSB237-AA001-DN06R USB PD sink controller was selected for USB-C power negotiation:

- **Minimalist configuration:** A single external capacitor ($C23 = 20\text{nF}$) on the VSET pin selects the target 20V voltage, eliminating the need for external resistors or programming.
- **Power capability:** Supports up to 20V/5A (100W), exceeding the 65W requirement.
- **Package:** DFN-6L (2mm $\times$ 2mm) for compact layout.

The HUSB237 negotiates with the USB-C PD source to request the voltage configured by the VSET capacitor; a 20nF capacitor selects 20V ([Hynetek Semiconductor, 2024](#)).

5.3 Microcontroller: STM32F405RGIx

The STMicroelectronics STM32F405RGIx was selected as mandated by SRS constraint SC-2 (“gimbal control hardware shall use an STM32-family microcontroller board supplied or approved by the McMaster Rocketry Team”). Key specifications:

- **Core:** ARM Cortex-M4 with FPU at up to 168MHz
- **Memory:** 1MB Flash, 192KB SRAM
- **Peripherals:** USART, timers, GPIO, SWD debug
- **Package:** LQFP-64 for hand-solderability

The MCU runs the `embassy-rs` async runtime for real-time servo control and UART communication ([STM, 2022](#)).

5.4 Inductor: FXL0630-15q-M

The Taiyo Yuden FXL0630-15q-M 15 μH inductor was selected for all three buck converters. Its 6.3mm $\times$ 6.0mm package provides current rating suitable for the 3A TPS54308 application. The 15 μH value balances ripple current and physical size at the 350kHz switching frequency.

5.5 Connectors

- **Jetson interface:** Samtec ESQ-106-33-L-D 6-pin header carries UART (RX/TX), 3.3V, and ground between the PCB and Jetson expansion header.
- **Servo outputs:** 3-pin 0.1" headers (signal, power, ground) for standard hobby servo connectors.
- **Jetson power:** 2-pin screw terminal for the 17V output to the Jetson's DC barrel jack.
- **USB-C:** HX TYPE-C 16P SMD 5A connector for PD input.
- **Debug:** SWD header for programming and debugging the STM32F405.

6 Circuit Design and Calculations

This section details the circuit implementation for each functional block, including component value calculations.

6.1 USB-C Power Delivery

The USB-C power input stage uses the HUSB237 PD controller to negotiate 20V from the connected power source. Figure 2 shows the implementation.

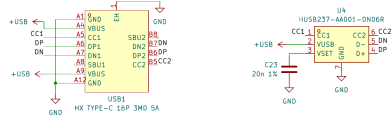


Figure 2: USB-C connector and HUSB237 PD controller schematic. The HUSB237 (U4) negotiates the PD contract; capacitor C23 (20nF) on VSET selects 20V target. USB1 is the Type-C receptacle.

The HUSB237 VSET pin uses an external capacitor to configure the target RDO (Request Data Object) voltage. The relationship between capacitance and target voltage is defined by the manufacturer; for the HUSB237, a 20nF capacitor selects 20V. This simple configuration eliminates the need for external resistors or I2C programming.

Decoupling capacitor C23 (20nF, 1% tolerance ceramic) connects between VSET and ground. A 1 μ F decoupling capacitor near the VBUS pin provides supply filtering. The CC1 and CC2 pins connect to the USB-C Configuration Channel lines for PD communication.

6.2 Servo Power Regulators

Two identical TPS54308 buck converters generate the 5.5V rails for the pan and tilt servos. Figure 3 shows one channel; the second channel is duplicated.

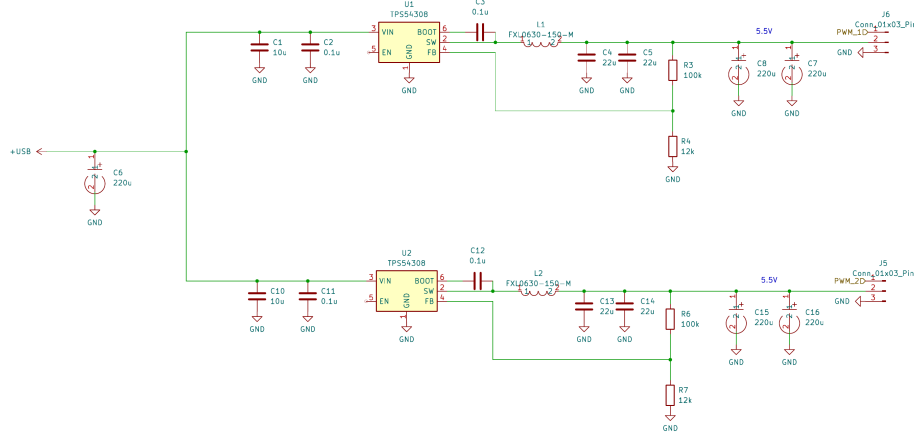


Figure 3: Servo power regulator schematic (one channel shown; two identical channels implemented). The TPS54308 (U2) steps down 20V to 5.5V with feedback divider R3/R4 and output filter capacitors C8, C7, C13, C14, C5.

6.2.1 Output Voltage Calculation

The TPS54308 regulates its output to maintain 0.6V at the FB (feedback) pin. The output voltage is set by a resistive divider:

$$V_{out} = V_{ref} \times \left(1 + \frac{R_{top}}{R_{bottom}}\right) \quad (1)$$

where $V_{ref} = 0.6V$ (internal reference).

For the servo rail ($V_{out} \approx 5.5V$):

$$R_{top} = R_3 = 100k\Omega \quad (2)$$

$$R_{bottom} = R_4 = 12k\Omega \quad (3)$$

$$V_{out} = 0.6 \times \left(1 + \frac{100k}{12k}\right) \quad (4)$$

$$= 0.6 \times (1 + 8.333) \quad (5)$$

$$= 0.6 \times 9.333 \quad (6)$$

$$= 5.6V \approx 5.5V \text{ (nominal)} \quad (7)$$

The 5.6V output provides margin above the nominal 5.5V to compensate for cable drops while remaining below the typical 6V maximum rating of hobby servos.

6.2.2 Component Selection

- **Inductor:** L1 (FXL0630-15q-M, 15 μ H) selected for 350kHz operation at 2A load.
- **Output capacitors:** C8/C7 (220 μ F electrolytic) provide bulk capacitance for transient servo loads; C13/C14 (22 μ F ceramic) filter high-frequency ripple.
- **Input capacitors:** C1 (10 μ F) and C2 (0.1 μ F) provide input filtering.
- **Boot capacitor:** C3 (0.1 μ F) between BOOT and SW pins provides gate drive voltage for the high-side MOSFET.

6.3 Jetson Power Regulator

A third TPS54308 buck converter generates the 17V rail for the Jetson Orin Nano. Figure 4 shows the circuit.

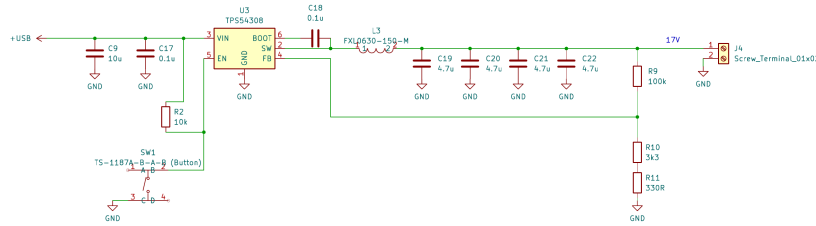


Figure 4: Jetson power regulator schematic. The TPS54308 (U3) steps down 20V to 17V with feedback divider R9/R10/R11. The EN pin is controlled by switch SW1 for power control. Output filtering uses four 4.7 μ F ceramic capacitors (C19-C22).

6.3.1 Output Voltage Calculation

The feedback divider uses a series combination for R_{bottom} to achieve a non-standard ratio:

$$R_{top} = R_9 = 100\text{k}\Omega \quad (8)$$

$$R_{bottom} = R_{10} + R_{11} = 3.3\text{k}\Omega + 330\Omega = 3.63\text{k}\Omega \quad (9)$$

$$V_{out} = 0.6 \times \left(1 + \frac{100\text{k}}{3.63\text{k}}\right) \quad (10)$$

$$= 0.6 \times (1 + 27.55) \quad (11)$$

$$= 0.6 \times 28.55 \quad (12)$$

$$= 17.13\text{V} \approx 17\text{V (nominal)} \quad (13)$$

The 17V output provides margin above the Jetson's minimum 7V input while remaining well below the maximum 20V rating ([NVIDIA, 2024](#)).

6.3.2 Enable Circuit

The TPS54308 EN (enable) pin controls regulator operation. Switch SW1 (TS-1187A-B-A-B tactile button) pulls the EN pin high when pressed, activating the regulator. This provides a soft-start capability for the Jetson power rail.

6.4 STM32 Microcontroller

The STM32F405RGIx microcontroller forms the core of the gimbal control system. Figure 5 shows the MCU and its supporting circuitry.

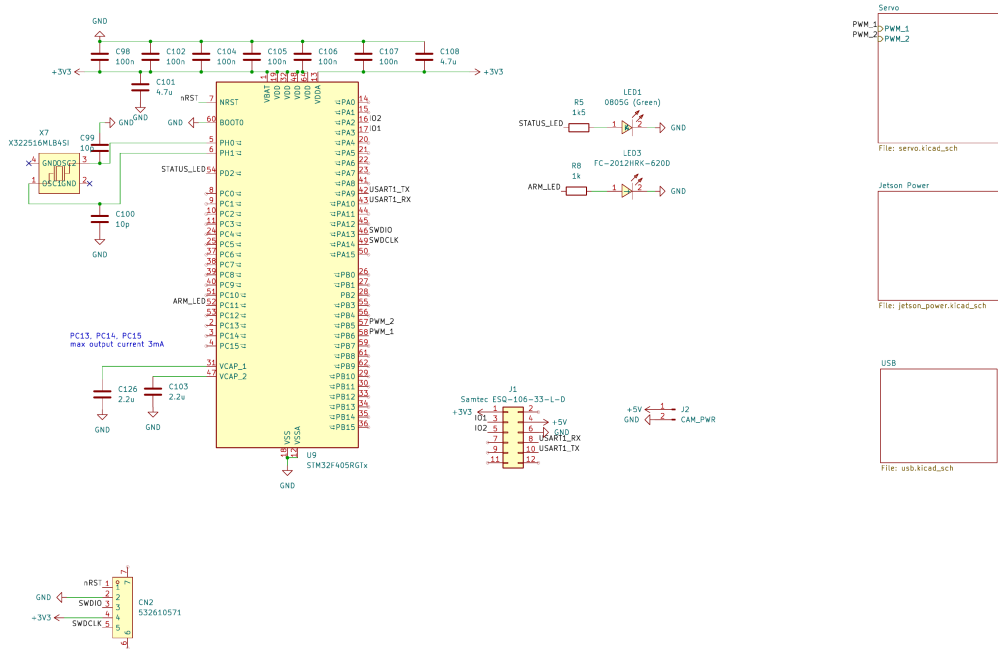


Figure 5: STM32F405 microcontroller schematic. The MCU (U9) includes decoupling capacitors (C98-C108), VCAP capacitors (C126, C103), 8MHz crystal (X7) with load capacitors, status LEDs, and the Samtec connector for Jetson interface.

6.4.1 Decoupling and Power

The STM32F405 requires extensive decoupling for stable operation:

- **VDD decoupling:** Five 100nF ceramic capacitors (C98, C102, C104, C105, C106) distributed near the VDD pins filter high-frequency noise.
- **Bulk decoupling:** One 4.7µF capacitor (C101) provides low-frequency energy storage.
- **VCAP capacitors:** Two 2.2µF capacitors (C126, C103) connect to the VCAP_1 and VCAP_2 pins for the internal voltage regulator.

6.4.2 Clock Source

An 8MHz crystal (X7, X322516MLB4SI) with 10pF load capacitors (C99, C100) provides the HSE (High-Speed External) clock. The internal PLL multiplies this to the system clock frequency (up to 168MHz).

6.4.3 Status Indicators

Two LEDs provide visual status indication:

- **STATUS_LED** (green, D1): Controlled via PD2; indicates system status/heartbeat.
- **ARM_LED** (red/amber, D3): Controlled via PC11; indicates armed/disarmed state.

Current-limiting resistors R5 (1k Ω) and R8 (1k Ω) set LED current to approximately 3mA.

6.5 Communication Interfaces

The PCB implements two primary communication interfaces between the STM32 and external systems.

6.5.1 UART to Jetson

USART1 on the STM32F405 provides the serial link to the Jetson Orin Nano:

- **Pins:** PA9 (USART1_TX), PA10 (USART1_RX)
- **Configuration:** 115200 baud, 8 data bits, no parity, 1 stop bit (8N1)
- **Protocol:** Gimbal Protocol as defined by McMaster Rocketry Team (command IDs 0x00–0x07)
- **Connector:** Samtec ESQ-106-33-L-D (J1) carries TX, RX, 3.3V, and GND

6.5.2 PWM to Servos

Software-generated PWM signals control the servos:

- **Tilt servo:** PB5, 50Hz update, 1040–2360 μ s pulse width, -90°to +90°range, reversed direction, 49°offset.
- **Pan servo:** PB6, 50Hz update, 1040–2360 μ s pulse width, -90°to +90°range, standard direction, -7°offset.

The pulse width is calculated as:

$$\text{pulse_width} = 1040 + \frac{\text{angle} + 90}{180} \times 1320 \quad \mu\text{s} \quad (14)$$

7 PCB Layout

Figure 6 shows the PCB layout for the “Tracking Cam Prototype 0.5” board.

7.1 Layout Considerations

- **Power traces:** High-current paths (20V input, 17V output, 5.5V outputs) use wide traces with thermal relief for component pads.
- **Ground plane:** A solid ground plane on the bottom layer provides low-impedance return paths and EMI reduction.
- **Decoupling placement:** Capacitors placed close to IC power pins minimize loop inductance.
- **Switching noise:** The TPS54308 switch node (SW) traces are kept short to reduce radiated EMI.
- **Connector placement:** USB-C, servo headers, and Jetson interface placed on board edges for cable management.
- **Mounting holes:** Four mounting holes with copper keepouts for mechanical attachment to the gimbal frame.

8 Conclusion

The “Tracking Cam Prototype 0.5” PCB successfully integrates the power and control functions required for the RoCam gimbal tracking system. The design delivers:

- **65W USB-C PD input** with 20V negotiation via the HUSB237 controller.
- **Three regulated rails:** 17V/2A (Jetson), 5.5V/2A (pan servo), 5.5V/2A (tilt servo).
- **STM32F405-based control** with UART to Jetson and PWM servo outputs.
- **Compliance with project requirements:** STM32-family MCU (SRS SC-2), Jetson Orin Nano support (SRS SC-3), and operating temperature range 0–45°C (SRS EPE-1).

All feedback resistor values have been calculated and verified against the TPS54308 0.6V reference voltage. The board is ready for assembly and integration testing with the RoCam system.

References

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- STM32F405/415, STM32F407/417, STM32F427/437 and STM32F429/439 Reference Manual*. STMicroelectronics, 2022. RM0090 Rev 20, accessed 2026-03-04.
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A Full Schematics

This appendix presents the complete schematic set for the Tracking Cam Prototype 0.5 PCB.

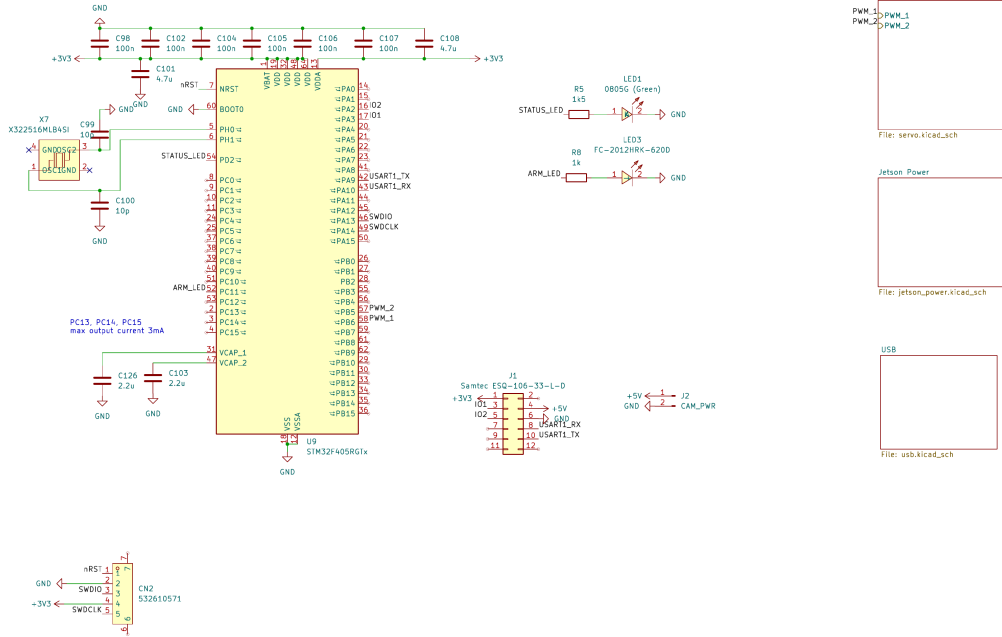


Figure 7: STM32F405 microcontroller and LED schematic (Sheet 1 of 4).

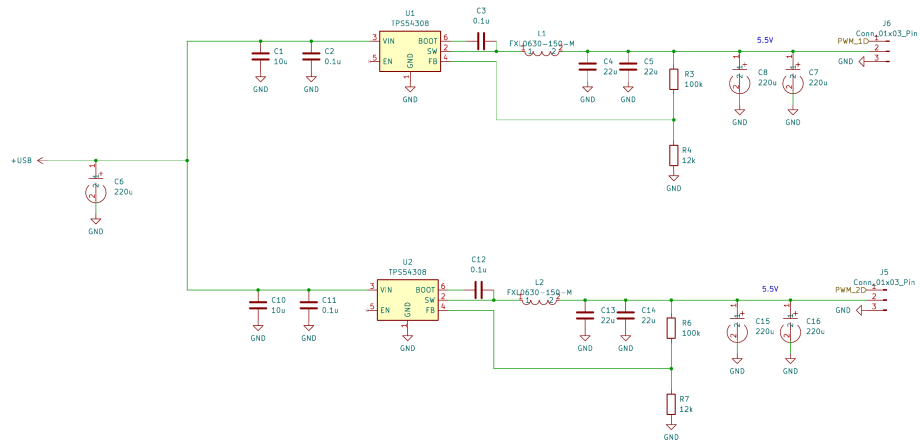


Figure 8: Servo power regulators schematic (Sheet 2 of 4).

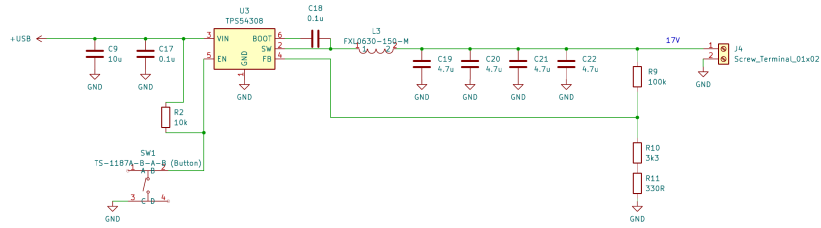


Figure 9: Jetson power regulator schematic (Sheet 3 of 4).

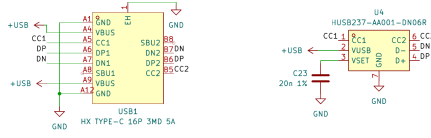


Figure 10: USB-C connector and PD controller schematic (Sheet 4 of 4).

B Bill of Materials

Table 2 lists the key components for the PCB assembly.

Table 2: Bill of Materials for Tracking Cam Prototype 0.5.

Ref	Description	Qty	Value/Part
U2, U3	Buck converter, 3A, SOT-23-6	3	TPS54308DDCT
U4	USB PD sink controller, DFN-6	1	HUSB237-AA001-DN06R
U9	MCU, ARM Cortex-M4, LQFP-64	1	STM32F405RGT6
L1, L2, L3	Power inductor, 15 μ H	3	FXL0630-15q-M
X7	Crystal, 8MHz, SMD	1	X322516MLB4SI
USB1	USB-C receptacle, 16P, 5A	1	HX TYPE-C 16P
J1	Header, 6-pin, 2.54mm	1	Samtec ESQ-106-33-L-D
J4	Screw terminal, 2-pin	1	Screw_Terminal_L01x02
SW1	Tactile switch, SMD	1	TS-1187A-B-A-B
D1, D3	LED, SMD 0603	2	Green/Red
C98-C108	Capacitor, ceramic, 100nF	5	100nF, 50V
C101	Capacitor, ceramic, 4.7 μ F	1	4.7 μ F, 16V
C126, C103	Capacitor, ceramic, 2.2 μ F	2	2.2 μ F, 16V
C99, C100	Capacitor, ceramic, 10pF	2	10pF, 50V
C23	Capacitor, ceramic, 20nF	1	20nF, 1%

Note: Additional passive components (resistors, capacitors) for the regulator circuits are not listed exhaustively. See schematics for complete component values.